

Blackout

System Operator • Calder Interconnection

15 DAYS • 61 STOPS • GRADE 12 • BLACKOUT

THE OPENING CARD

Four million people are supplied from the Calder network, and the whole of it runs on one number. Mains frequency has to stay within half a hertz of fifty. Drift further and generators start tripping off, which takes the rest of the system with them. You are the system operator on nights. The power stations belong to companies you can instruct under contract and cannot order about. Demand rises and falls with whatever four million people happen to be doing. Holding the two together, second by second, is the job. A fault takes seconds. Getting a system back after one has gone down takes days, and the hospitals on this network hold about eight hours of their own generation.

Enough in the tank

BEFORE THE DAY — GREET

The shift changes at seven and nobody has met you

Two crews work this station and they hand over at seven. Farrow, the metering and standards engineer, wants you known to both before anything goes wrong, because at four in the morning a voice on a radio is either somebody you have met or somebody you are going to argue with. Get round the yard and put a name to most of the shift before the handover is done.

PLAN CARD 119W

Monday, and the forecast for tomorrow's evening peak has moved four degrees colder. Sten Lindgren says the number has not changed much but the width has, and the width is what the reserve has to cover. Rafael Alvarez has the storage figures from the weekend and they are worse than the brochure. Dolores Reyes has to say tonight how much generation is held back and how much is sold. Today you work out what the battery will really give back, decide why the transmission side runs at high voltage at all, and choose what evidence is worth buying before the peak. Hold too much back and it is wasted. Hold too little and the first cold evening finds out.

BRIEFING 12W

Tomorrow's peak is colder than the forecast the reserve was set against.

WHAT YOU DECIDE 12W

Decide what the system is holding back, and whether it is enough.

1.1 What comes out against what went in

LOAD & FORECASTING · A ROOM · BALLPARK

SCENE 39W

The battery charged overnight on Saturday and discharged into Sunday's peak. Alvarez, the storage engineer, has both energy totals and the brochure figure. The three numbers do not agree. The weekend log is open on the desk beside them.

GUIDE 67W

Four numbers, and two of them are the power rating and the rated duration. Those say how fast and for how long, not how much came back. Ask of each whether it is an energy or a rate. A round trip is what went in against what came out, over a whole cycle. Three numbers on the desk disagree, and the brochure figure is one of them.

ASK 7W

Estimate the round-trip efficiency of the store.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

storing energy and recovering it both lose some of it · electrical power and energy over time — taken as read

VERDICT 76W

Everything that moves energy in and out loses a share of it, in the converter, in the chemistry and in keeping itself cool. Round-trip efficiency is the plain ratio of what came out to what went in, over a whole cycle. It is the only figure that tells you what the store is worth as reserve. A power rating is a different claim: it says how fast, not how much. Duration is a third one again.

TAKEAWAY 12W

Round-trip efficiency is measured on the energy, not on the power rating.

1.2 The reason the wires are not at wall voltage

GENERATION · GENERATION HALL · A PERSON · CHOICE

SCENE 31W

A visitor asks Nadia Haddad, the generation lead, why the machine makes power at 20000 volts and the line outside runs at 400000. The room is waiting for a written instruction.

GUIDE 64W

All four options pair a change in current with a change in loss. Two of them get the current right and disagree about the heat. Ask what the loss depends on, and what power it depends on it to. Current falls in proportion to voltage. Heat follows the square of current. Work both halves before choosing, because three of these get one half right.

ASK 17W

For the same real power, what happens when voltage is stepped from 20 kV to 400 kV?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a transformer changes voltage in proportion to its turns · electrical power and energy over time — taken as read · faraday's law and electromagnetic induction — taken as read

VERDICT 78W

For a given real power, raising voltage allows the same power to move with less current. Resistive loss then falls with current squared. Halving current cuts that loss to one quarter. A transformer makes the trade possible by changing voltage in proportion to turns while transferring nearly the same power, apart from losses. That is why long-distance transmission uses high voltage despite the harder

insulation problem. The cost of insulation buys a much larger saving in conductor heating.

TAKEAWAY 18W

The same power can be delivered in more than one way, and the ways are not equally wasteful.

1.3 What a node has to add up to

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 31W

The busbar meter reads 900 A out. Reyes wants to know what that single figure rules out about the two circuits underneath it. A printed network diagram lies on the desk.

GUIDE 63W

All four options say something about two parallel circuits. They differ in whether they need one equation or two. Ask of each whether the meter reading alone could establish it. Charge does not pile up at a junction, so the branch currents have to add to the total. That fixes their sum and nothing else. Two of these options confuse parallel with series.

ASK 12W

What does the 900 A total tell you about the 2 circuits?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

charge does not accumulate at a junction · ohm's law and resistive networks — taken as read · charge, current, voltage and resistance — taken as read

VERDICT 85W

Charge does not build up at a junction, so current entering a node must equal current leaving it. Kirchhoff's current law gives that conservation rule. It fixes the sum of the branch currents without telling you how the total divides. The split needs another relation, such as the branch impedances. Assuming an even split is wrong when the paths differ. Assuming each branch carries the full total confuses parallel and series circuits. One equation constrains two unknown branch currents; it does not determine both alone.

TAKEAWAY 14W

A conservation law bounds the parts of a system without identifying any of them.

END OF THE DAY 19W

A reserve is bought with something, and the argument is always about how much uncertainty it has to cover.

Three things at once

BEFORE THE DAY — TRIAL

Walk the yard before you touch anything

Nobody works on a switching station they have not walked. Reyes, the shift supervisor, takes new engineers round the yard once, in daylight, before the first shift: the control room, the relay house, the transformer bays, the metering hut. Every one of them is somewhere you will be sent this fortnight, and the map on the plan card means nothing until you have stood in the places it names.

PLAN CARD 119W

Tuesday, 16:40, and 3 things arrive together. A cable fault has taken a feeder with 3,300 households on it. Ravi Lindgren's forecast is running 200 megawatts under actual and still climbing. And the gas machine that has carried the evening peak all month has to come off at 1800 for a fuel limit Amira Haddad flagged this morning and nobody built a plan around. Dolores Reyes has 1 control room, 1 crew on shift and about 80 minutes. Today you decide what gets attention first, work out what the corridor is costing at the higher flow, and put the evening's moves in an order that does not undo itself. None of these is new. All three at once is.

BRIEFING 17W

A fault, a forecast miss and a plant coming off at the same hour, and one room.

WHAT YOU DECIDE 11W

Rank three simultaneous problems, and hold the reasoning while they interfere.

2.1 Which of the three cannot wait

SYSTEM OPERATIONS • A ROOM • DELEGATE

SCENE 35W

Reyes has three live problems on the board and only one attention channel she can keep herself. Aaron Whitlock is ready to take a handoff, but she refuses any instruction that amounts to “watch it.”.

GUIDE 60W

Three live problems and one attention channel you can keep. For each of the other two, name an owner, the first real action they will take, and the reading or event that brings the problem back to you. Reyes will not accept anything that amounts to 'watch it'. Then take the watch on the one that needs your own judgement.

ASK 11W

What do you keep, and what exactly do you hand off?

BEHIND THE BUTTON 138W

Why 'watch it' fails. An owner who has been told to observe will come back at the moment they need instructions, which is the moment you are busiest. A first action is what lets somebody hold a problem without you, and naming it is the difference between delegating and deferring.

What a return condition is for. It is not a promise to check in. It is the threshold at which the problem stops being theirs, agreed in advance, so nobody has to judge in the moment whether this is bad enough to interrupt you.

How to choose what to keep. Compare the three on how fast they are getting worse, on whether they are already contained, and on what happens if nobody acts. The one that needs judgement rather than procedure is the one you cannot hand over.

TAKES AS READ

a stable problem can be severe without becoming more urgent each minute • reserve actions need lead time before the peak

VERDICT 86W

The cable outage is loud but stable: customers are already off and a crew is moving. The fuel-limited machine has a known time and quantity, so another operator can prepare its replacement. The forecast error is different. It is still growing, and reserve procurement needs lead time before the evening peak. Every update that passes without action removes options that cannot be recovered later. Delegation is therefore not “watch it.” Each handoff needs an owner, a first action and a condition that brings the problem back.

TAKEAWAY 22W

Urgency comes from how fast a condition is worsening, how reversible it is, and whether somebody else can own the next move.

2.2 Loss when the load turns up anyway

TRANSMISSION & PROTECTION · A PERSON · BALLPARK

SCENE 36W

With the feeder out, flow has redistributed and the corridor is carrying more. Novak wants the heat figure before he decides whether the evening needs a switching change. A printed network diagram lies on the desk.

GUIDE 66W

Five numbers, and two of them are the current an hour ago and the conductor temperature. Neither belongs in the loss. Ask of each number whether the heat depends on it. And note the square. Current has risen about eleven per cent, and the heat rises by nearly a quarter. The rating is thermal, so that is a bigger bite of the margin than it looks.

ASK 10W

Estimate the power now lost as heat in the corridor.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

loss in a conductor rises with the square of the current · ohm's law and resistive networks — taken as read · electrical power and energy over time — taken as read

VERDICT 72W

The same loss relation now gives a worse answer. Resistive heating rises with the square of RMS current, across all three phases. Current has risen by about 11%, from 1,150 A to 1,280 A. The heat rises by about 24%, from roughly 17 MW to 21 MW. That matters because the conductor rating is thermal. A small-looking increase in current can therefore use a much larger fraction of the available temperature margin.

TAKEAWAY 17W

When a quantity enters squared, a small change in it is a large change in the result.

2.3 An order that survives being carried out

LOAD & FORECASTING · A ROOM · SEQUENCE

SCENE 32W

Lindgren has four moves for the evening and no view on the order. Two of them change what the others are worth, and one of them cannot be reversed inside the shift.

GUIDE 62W

All four moves are available right now, so this is not a clock. Ask of each two things. What does it improve about the numbers you are working from? And how easily can it be taken back? One of these sizes every other figure in the shift. One of them costs money rather than customers, and one takes real minutes to load.

ASK 10W

Order the evening's moves so none of them undoes another

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

some actions change the value of the actions that follow them

VERDICT 73W

Correct the forecast first. Every other number is sized against it, and a wrong demand figure means doing the right thing to the wrong amount. Bring reserve forward next. It is reversible, it costs money rather than customers, and it covers the machine coming off at 1800. Then move the replacement plant, which takes real minutes to load and has to be underway before the fuel limit bites. Reconfiguring the network is last.

TAKEAWAY 15W

Do the moves that improve your information first and the moves you cannot undo last.

Ranking is not about which problem is worst but about which one gets worse fastest and which one is cheapest to remove.

The number on the wall

BEFORE THE DAY — FOLLOW

The switching route, walked once by somebody who knows it

Obi, the distribution lead, walks the restoration route the way it has to be walked, in the order the isolations happen, and she will not do it twice. Stay with her. Too far back and you lose which bay she turned into; too close and you are standing in front of the cubicle she is reading. She stops at each isolation point long enough to look at it, and that is where the walk is actually learned.

PLAN CARD 148W

Wednesday, 04:12, and the largest unit on the Calder system has tripped without warning. Frequency fell and then steadied somewhere below where it began, which Dolores Reyes says matters more than the fall itself — a system that stops falling has found a new balance, and the new balance is short. Nadia Haddad has machines that can make up the shortfall but not in the next two minutes. June Farrow wants the records checked before anybody calls it a lost unit, and Reyes is not waiting: a system this far short does not get more certain while you read about it. This morning Reyes is right. Today you read what the frequency is telling you, work out how much generation actually went missing, and decide what to ask for first. Read it as a fault on one machine and you will spend the morning fixing the wrong thing.

BRIEFING 16W

A unit tripped at 04:12 and the frequency has not come back to where it started.

WHAT YOU DECIDE 14W

Establish what the frequency is actually telling the room before anybody acts on it.

3.1 What a frequency trend is evidence of

SYSTEM OPERATIONS · A ROOM · PROTOCOL

SCENE 36W

The wall shows frequency dropping for eight seconds, then flattening low. Aaron Whitlock, the assistant operator, has the log open. Two people have already named a cause, and he wants a statement the record can defend.

GUIDE 62W

Four observations, and the right column says what each one is evidence of. Pair them by asking one question. Is this quantity system-wide, or local to a place? Frequency is one number for the whole interconnection. Voltage is not. That difference is what lets a trend say whether something happened, while a voltage says where. Two people have already named a cause.

ASK 10W

From a falling trend to a statement about the system

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an alternating supply has a frequency measured in cycles per second · synchronous machines and synchronisation — taken as read · electrical power and energy over time — taken as read

VERDICT 80W

System frequency reflects the balance between real power entering the interconnection and real power being demanded. After a sudden generation loss, stored kinetic energy in synchronized machines supplies the deficit briefly, so frequency begins to fall. The initial rate of change depends on both the power mismatch and that stored energy. A later flattening can be governor response at a

lower frequency, not full recovery. Frequency is therefore strong evidence of a system-wide imbalance, while voltage is much more local.

TAKEAWAY 20W

Frequency is a system-wide balance signal; its rate of change depends on both the power mismatch and stored kinetic energy.

3.2 How much power actually left the system

GENERATION • GENERATION HALL • A PERSON • BALLPARK

SCENE 33W

Haddad, the generation lead, needs a number before starting anything. The only clean evidence is the first two seconds of the trend and the stored-energy figure for the machines that were online overnight.

GUIDE 66W

Five numbers, and two of them belong to the aftermath rather than the first instant. The settled deviation is what the governors did afterwards. The generation online is the denominator of a different question. Ask of each number whether it describes the first two seconds. A figure that is right about a later moment is wrong here, and the control room acts on whatever comes out.

ASK 10W

Estimate the generation lost at the moment of the trip.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate of change can be read off the steep part of a trend • synchronous machines and synchronisation — taken as read • electrical power and energy over time — taken as read

VERDICT 71W

Immediately after a generator trip, governor and reserve actions have not yet changed the power balance. Stored kinetic energy supplies the shortfall for the first moments, so frequency moves. With stored kinetic energy E_k , nominal frequency f_0 and measured rate df/dt , the imbalance is $\Delta P = 2E_k |df/dt| / f_0$. Using 25,000 MW·s, 50 Hz and 0.30 Hz/s gives about 300 MW. This estimate is most useful before slower controls begin reshaping the trace.

TAKEAWAY 15W

The initial rate of frequency change measures the size of the imbalance, given the inertia.

3.3 The order the records claim things happened in

METERING & STANDARDS · A ROOM · TRACE

SCENE 31W

June Farrow, the metering engineer, has three records of the same eight seconds. They disagree about what happened first. One record was stamped by a clock nobody has checked since spring.

GUIDE 56W

Open each record to see which clock and which physical input it depends on. Keep the ones that would still stand if one clock were wrong, and untick the rest. Then name the shared source behind the false early timestamps. Three records disagreeing about the order is evidence about the records, not only about the event.

ASK 16W

Which records still support the event order, and what shared source explains the false early timestamps?

BEHIND THE BUTTON 141W

Why a clock is part of the measurement. A record's timestamp is as good as the clock that made it, and a clock nobody has checked since spring can be seconds out. Every event stamped by it moves together, which is exactly what makes a false ordering look consistent.

How to tell dependence from agreement. Two records agreeing prove nothing if both were stamped by the same clock. What matters is whether a record's timing came from somewhere else — a synchronised source, a physical input with its own time base — because only those constrain the order independently.

What the order is for. Sequence-of-events analysis is how a blackout is explained: what tripped first decides whether this was a protection failure, an operator action or a cascade. Getting the order wrong produces a confident and wrong account of the night.

TAKES AS READ

a timestamp combines a measured event with a clock reference · two channels can agree because they share the same source · faraday's law and electromagnetic induction — taken as read · RMS against peak and average — taken as read

VERDICT 80W

A timestamp contains a physical event and a clock claim. Those can fail separately. Two relay-derived records can agree because they share one internal clock, so their agreement is not two independent measurements. The GPS disturbance recorder uses a different time source and preserves the physical ordering. The slower SCADA record can still confirm breaker state, even though its two-second scan cannot resolve sub-second order. Dependency tracing keeps useful evidence instead of discarding every channel touched by the bad clock.

TAKEAWAY 14W

Agreement is independent evidence only when the channels do not share the same dependency.

3.4 What comes out against what went in

LOAD & FORECASTING · A ROOM · CALLBACK · BALLPARK

SCENE 39W

The battery charged overnight on Saturday and discharged into Sunday's peak. Alvarez, the storage engineer, has both energy totals and the brochure figure. The three numbers do not agree. The weekend log is open on the desk beside them.

GUIDE 67W

Four numbers, and two of them are the power rating and the rated duration. Those say how fast and for how long, not how much came back. Ask of each whether it is an energy or a rate. A round trip is what went in against what came out, over a whole cycle. Three numbers on the desk disagree, and the brochure figure is one of them.

ASK 7W

Estimate the round-trip efficiency of the store.

BEHIND THE BUTTON 106W

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TAKES AS READ

storing energy and recovering it both lose some of it · electrical power and energy over time — taken as read

VERDICT 76W

Everything that moves energy in and out loses a share of it, in the converter, in the chemistry and in keeping itself cool. Round-trip efficiency is the plain ratio of what came out to what went in, over a whole cycle. It is the only figure that tells you what the store is worth as reserve. A power rating is a different claim: it says how fast, not how much. Duration is a third one again.

TAKEAWAY 12W

Round-trip efficiency is measured on the energy, not on the power rating.

END OF THE DAY 20W

Frequency is not a reading from a place; it is the whole system's supply and demand, arriving as a number.

What the line can carry

PLAN CARD 129W

Thursday, and the replacement generation is coming from the wrong side of the system, which means it is arriving down two circuits instead of three. Piotr Novak has a corridor sitting above its continuous rating and a relay that will take the decision away from the room in about 40 minutes if nobody else does. The 40 minutes comes off a thermal record June Farrow has not verified, and she says so. Thabo Dube is in the yard waiting for a switching instruction. Today you rate the line honestly, work out where the flow goes if it trips, and decide whether to move generation or accept the risk. A thermal rating is not a cliff, and treating it as one is how a warm line becomes two cold ones.

BRIEFING 14W

With the unit off, the flow has redistributed and one corridor is running warm.

WHAT YOU DECIDE 11W

Decide which circuit takes the flow once one path is gone.

4.1 What you do before a relay does it for you

TRANSMISSION & PROTECTION · A ROOM · TRIGGER

SCENE 34W

The corridor is at 108% of continuous rating and rising slowly. Novak, the transmission engineer, wants a response he can hand to Dube, the switching operator, before the relay makes the choice for them.

GUIDE 75W

One rule, written before the trend arrives and fixed once you release it. Moving 200 MW off the corridor takes six hours from the moment it is called, and at 120 % of continuous rating the protection takes the line out itself. So the line goes back from 120 by however far the loading climbs in those six hours — and not so far back that you are redispatching a corridor which has shown nothing.

ASK 20W

At what loading do you commit the redispatch, given the move takes six hours and protection acts at 120 %?

BEHIND THE BUTTON 140W

Why act before the relay. Protection will disconnect the corridor when its limit is reached, and it will do so without regard to what else that leaves connected. Anything you do before then is a controlled choice; anything after is a consequence.

Why two thresholds rather than one. Redispatching generation takes minutes and shedding load takes seconds. A single threshold set for the slower action fires far too early for the faster one, and set for the faster one leaves no time for the slower. The lead time is the reason there are two numbers.

What Dube needs from this. A switching operator acting at three in the morning needs a rule, not a judgement. Numbers agreed in advance are what make the night's actions defensible afterwards, and what stop the decision being made by whoever happens to be awake.

TAKES AS READ

continuous and emergency ratings are different limits · redispatch is slower than automatic or operator load shedding · charge, current, voltage and resistance — taken as read · ohm's law and resistive networks — taken as read

VERDICT 101W

An emergency rating is a time-limited thermal allowance, not permission to wait until the relay acts. Redispatch takes six hours from the moment it is called, so the loading at which it is committed has to sit back from the 120 per cent ceiling by however far the corridor climbs in those six hours. That is what makes the number smaller than the number the ceiling suggests. Writing the rule before the trend appears is what keeps hindsight from turning into policy, and a threshold that is physically sensible but fires after its action can still arrive is a failed rule.

TAKEAWAY 20W

A protection plan must be written before the number moves, with earlier thresholds for actions that take longer to arrive.

4.2 The power that never arrives

LOAD & FORECASTING · A PERSON · BALLPARK

SCENE 32W

Sten Lindgren, who forecasts demand, wants the loss figure before the morning call. The argument about moving generation is really an argument about waste. The yard radio is still live beside them.

GUIDE 61W

Five numbers, and two of them are the voltage and the power factor. Neither appears in a resistive loss. Ask of each number whether the heat depends on it. Heat comes from current and resistance, and it is paid once per phase. Note the square while you are choosing. Ten per cent more current is about twenty-one per cent more heat.

ASK 10W

Estimate the power lost as heat in the loaded corridor.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current through a resistance dissipates power as heat · charge, current, voltage and resistance — taken as read

VERDICT 81W

Resistive loss in one phase is the RMS current squared times that phase resistance. A balanced three-phase corridor has three such conductors, so the total is three times $I^2 R$. The square is the key. 10% more current means about 21% more heat. High transmission voltage helps because the same real power can be moved with less current. Lower current then cuts resistive loss sharply, which is why a modest overload can consume thermal margin faster than its percentage suggests.

TAKEAWAY 18W

Loss rises with the square of current, which is why a small overload costs more than it looks.

4.3 What the path opposes at 50 hertz

SYSTEM OPERATIONS • A ROOM • BALLPARK

SCENE 43W

Reyes has the switching order for the second circuit in front of her. Novak has read the path's own figures out of the records: 3.1 ohms of resistance, and 12.4 ohms of reactance at 50 hertz. She wants one number before she signs.

GUIDE 85W

Five numbers, and only two of them are properties of the path. Ask of each whether it describes what the path is, or what is being pushed through it. Resistance and reactance are both measured in ohms and both oppose the current, but they do it at different moments in the cycle, so they do not add end to end. Combine them the way you would two sides of a right-angled triangle, and notice before you commit how little the smaller one moves the answer.

ASK 9W

Estimate what the corridor path opposes at 50 hertz.

BEHIND THE BUTTON 190W

Why the two do not simply add. Resistance turns electrical energy into heat and opposes the current itself. An inductance stores energy in a magnetic field and hands it back a quarter of a cycle later, so what it opposes is the change of current. The two oppositions peak at different instants, and adding them arithmetically would count a maximum that never happens.

Why an overhead path is mostly reactance. A long circuit is a large loop of wire, and its inductance grows with the area that loop encloses. At the spacings a 400 kV tower needs, that inductance dominates: on this path the reactance is four times the resistance, so the resistance contributes about three per cent of the total, and heat is a small part of the story.

What the frequency has to do with it. Reactance is proportional to frequency, so a figure like this one is only true at the frequency it was measured at. Resistance is nearly a property of the metal; reactance is a property of the metal and the cycle together, which is why a record of it has to say 50 hertz.

TAKES AS READ

an alternating current has a frequency, and this system runs at 50 hertz • the hypotenuse of a right-angled triangle is the square root of the sum of the squares • aC waveforms: frequency, period and phase — taken as read • capacitance, inductance and stored energy — taken as read • rMS against peak and average — taken as read

VERDICT 84W

Resistance opposes the current itself and turns part of it into heat. Reactance opposes the change of current, because an inductance answers a rising current by pushing back, and it does that a quarter of a cycle away from the resistance. Two oppositions that peak at different moments cannot be added end to end; they

combine as the two shorter sides of a right-angled triangle. Here the reactance is four times the resistance, so the resistance moves the total by about three per cent.

TAKEAWAY 22W

Two oppositions that peak a quarter of a cycle apart combine at right angles, so the larger of them dominates the total.

4.4 The reason the wires are not at wall voltage

GENERATION · GENERATION HALL · A ROOM · CALLBACK · CHOICE

SCENE 31W

A visitor asks Nadia Haddad, the generation lead, why the machine makes power at 20000 volts and the line outside runs at 400000. The room is waiting for a written instruction.

GUIDE 64W

All four options pair a change in current with a change in loss. Two of them get the current right and disagree about the heat. Ask what the loss depends on, and what power it depends on it to. Current falls in proportion to voltage. Heat follows the square of current. Work both halves before choosing, because three of these get one half right.

ASK 17W

For the same real power, what happens when voltage is stepped from 20 kV to 400 kV?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a transformer changes voltage in proportion to its turns · electrical power and energy over time — taken as read · faraday's law and electromagnetic induction — taken as read

VERDICT 78W

For a given real power, raising voltage allows the same power to move with less current. Resistive loss then falls with current squared. Halving current cuts that loss to one quarter. A transformer makes the trade possible by changing voltage in proportion to turns while transferring nearly the same power, apart from losses. That is why long-distance transmission uses high voltage despite the harder

insulation problem. The cost of insulation buys a much larger saving in conductor heating.

TAKEAWAY 18W

The same power can be delivered in more than one way, and the ways are not equally wasteful.

END OF THE DAY 20W

A rating is a time and a temperature, not a wall, and what matters is where the flow goes next.

The last few kilometres

PLAN CARD 106W

Friday, and the transmission side is steady, but 11 distribution feeders are still dead. Chinelo Obi has 3 crews and 14000 households in the dark. One of her circuits carries a dialysis clinic. Another carries 1000 homes. Ewa Kowalczyk is at the first fault already, and it is a tree. Today you work out why the far end of a long feeder sits below limit, decide what sets the current a conductor may carry, and choose which circuit gets the first crew. Every hour of this is somebody sitting in the cold. The order you choose is the order an inquiry will read back to you.

BRIEFING 17W

11 feeders are still dead and the crews can only be in 1 place at a time.

WHAT YOU DECIDE 12W

Get customers back on in an order that can be defended afterwards.

5.1 What is common and what divides

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · PROTOCOL

SCENE 30W

Obi, the distribution supervisor, is training a new starter. Four circuit sketches are on the whiteboard, and he wants each matched to the quantity that stays common across that arrangement.

GUIDE 61W

Four arrangements, and the right column names what stays common in each. Pair them by asking how many paths the current has. One path means one current and a divided voltage. Two nodes shared means one voltage and a divided current. Then ask what adding a branch does to the total. Getting this wrong sizes a fuse for the wrong current.

ASK 7W

Match each connection to what it shares

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a circuit can be connected end to end or side by side · kirchhoff's laws and nodal analysis — taken as read

VERDICT 86W

Series and parallel circuits differ by what stays common. In series there is one path, so every element carries the same current. The supply voltage divides across the elements. In parallel every branch connects to the same two nodes, so each branch has the same voltage. The current divides instead. Adding a parallel branch lowers the equivalent resistance and raises total current. Adding a series resistance raises the equivalent resistance and lowers the loop current. Mixing these rules leads directly to wrong fuse and conductor sizing.

TAKEAWAY 16W

Series shares the current and divides the voltage; parallel shares the voltage and divides the current.

5.2 What actually limits a conductor

TRANSMISSION & PROTECTION • A PERSON • PROTOCOL

SCENE 35W

Novak is asked to allow more current down a circuit for the afternoon. He wants each reason for saying no matched to the thing it is actually about. The shift log is open beside them.

GUIDE 63W

Four observations, and each is limited by a different thing. Ask of each whether it is set by current, by voltage, or by geometry. Heat comes from current and does not care about voltage. Insulation cares about voltage and not about current. A volt drop is a customer problem. One of these four is where a heat limit turns into a safety one.

ASK 7W

Match each observation to what it constrains

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

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TAKES AS READ

a conductor warms up as current passes through it • electrical power and energy over time — taken as read

VERDICT 83W

Two different limits get confused. Heat comes from current, and it does not care what voltage the conductor is at. The insulation and the clearances care about voltage, and they do not care what current is flowing. So a circuit can be at nominal voltage and still be over its thermal limit, which is the case in front of the room. Volt drop is a third thing again, a consequence of current, and it is a customer problem rather than a safety one.

TAKEAWAY 14W

Current sets the heat, voltage sets the insulation, and the two limits are separate.

5.3 The order somebody will read back to you

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 39W

3 crews, 11 dead feeders, and a list in front of Reyes that mixes household counts with 1 clinic and 1 pumping station. The control-room phone is already ringing.. The shift log and network board are open beside them.

GUIDE 61W

Four feeders, and household count is only one of two things that matter. Ask of each what waiting costs, and how long the work takes. A one-hour job frees the crew for the next fault this shift. A four-hour job does not. One of these is measured in treatments rather than in inconvenience, and one is already on a standby supply.

ASK 7W

Which feeder should take the first crew?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a crew can only work one fault at a time

VERDICT 75W

Two things decide this, and household count is only one of them. The first is what waiting costs, and for a dialysis clinic it is measured in treatments rather than in inconvenience. The second is how long the work takes, because a one-hour job frees the crew to take the next fault the same shift. The clinic circuit is worst on the first and best on the second, so it goes first on both counts.

TAKEAWAY 17W

The first crew goes where the consequence of waiting is worst, not where the count is biggest.

5.4 An order that survives being carried out

LOAD & FORECASTING · A ROOM · CALLBACK · SEQUENCE

SCENE 32W

Lindgren has four moves for the evening and no view on the order. Two of them change what the others are worth, and one of them cannot be reversed inside the shift.

GUIDE 62W

All four moves are available right now, so this is not a clock. Ask of each two things. What does it improve about the numbers you are working from? And how easily can it be taken back? One of these sizes every other figure in the shift. One of them costs money rather than customers, and one takes real minutes to load.

ASK 10W

Order the evening's moves so none of them undoes another

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

some actions change the value of the actions that follow them

VERDICT 73W

Correct the forecast first. Every other number is sized against it, and a wrong demand figure means doing the right thing to the wrong amount. Bring reserve forward next. It is reversible, it costs money rather than customers, and it covers the machine coming off at 1800. Then move the replacement plant, which takes real minutes to load and has to be underway before the fuel limit bites. Reconfiguring the network is last.

TAKEAWAY 15W

Do the moves that improve your information first and the moves you cannot undo last.

Restoration is a series of choices about people, made with instruments that only see the substation.

Is that number the number

BEFORE THE DAY — HUNT

Six earths, and the log says five

Before the line can be re-energised every temporary earth has to be off. The log says five were applied. Lindgren, the load forecasting analyst, counted six going on. Until all of them are found and accounted for nothing gets switched in, because an earth left on a line that is made live is a fault the size of the station. Find them.

PLAN CARD 134W

Monday, and the account of Wednesday night has to be written down. Dolores Reyes wants it filed today so the week can close. June Farrow will not sign it while two of its figures come from instruments last calibrated in the spring. Nadia Haddad needs a machine back on the bars before the evening peak, and there is a right order for that which is not the order anybody is in a hurry to follow. Today you work out what the system was actually delivering, put the steps of synchronising a machine in the order physics requires, and settle an argument between a meter and an oscilloscope. This time Farrow is right, and the delay is the cheapest thing in the week: a number in a report outlives everybody who knew what it meant.

BRIEFING 17W

The report is due and two of the figures in it come from instruments nobody has checked.

WHAT YOU DECIDE 11W

Establish which readings the incident account is allowed to rest on.

6.1 What three phases actually add up to

METERING & STANDARDS · A ROOM · BALLPARK

SCENE 36W

Farrow has line voltage, line current and a power factor from the night of the event. The report claims a figure for delivered power that nobody has reproduced. A printed network diagram lies on the desk.

GUIDE 71W

Six numbers, and two of them are traps that look right. There are three phases, and the factor is not three. And the sine of the angle is not what selects real power. Ask of each number what definition it belongs to. A voltage measured between two phases differs from one measured to neutral, and that is where the factor comes from. A report already claims a figure nobody has reproduced.

ASK 8W

Estimate the real power the circuit was delivering.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a three-phase supply carries power on three conductors at once · electrical power and energy over time — taken as read

VERDICT 72W

The tempting mistake is to multiply by three because there are three phases. The factor is root three instead. Line voltage is measured between two phases, not from one phase to neutral. Those voltage definitions differ by root three, so the line-quantity power formula carries that factor. Power factor then selects the in-phase current that produces real power. Using three instead of root three overstates the answer enough to change the report.

TAKEAWAY 16W

Three-phase power from line quantities carries a factor of root three, and it is not three.

6.2 Four things that have to agree before the breaker closes

GENERATION • GENERATION HALL • A PERSON • SEQUENCE

SCENE 32W

Haddad has a machine at rest and a system running at nominal. She wants the sequence written down before anybody is standing at the panel. The next operating call is minutes away.

GUIDE 61W

All four of these happen at the panel, so this is not a matter of what is available. Ask of each what mismatch it removes, and what that mismatch would do if the breaker closed with it still there. A speed difference and a voltage difference cause different damage. One of these four is the consequence of the others being right.

ASK 10W

Put the steps of synchronising in the order physics requires

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a generator joined to a live system is forced to turn at the system's speed • aC waveforms: frequency, period and phase — taken as read • faraday's law and electromagnetic induction — taken as read

VERDICT 80W

Each step removes a different mismatch before the breaker closes. Match frequency first, because a speed difference forces a violent torque when the machine is tied to the grid. Match voltage next to avoid a large reactive current. Then wait for phase angle to approach zero as the two waveforms drift past each other. The breaker closes only when frequency, voltage and phase are all inside their allowed windows. Closing earlier turns a synchronization error into electrical and mechanical stress.

TAKEAWAY 11W

Speed, then voltage, then phase angle, and only then the breaker.

6.3 Two instruments, two quantities

TRANSMISSION & PROTECTION · A ROOM · CHOICE

SCENE 35W

An oscilloscope on the busbar shows a peak of 325 volts. The panel meter beside it reads 230. Both were calibrated last month and neither has moved. The yard radio is still live beside them.

GUIDE 63W

Four relations, and both instruments are in calibration. So the question is what each number is a summary of. An alternating voltage has no single value. One instrument reports the highest the waveform reaches. The other reports the steady voltage that would heat a resistor the same. Ask which way the conversion runs before choosing, because two of these options have it backwards.

ASK 10W

Which numerical relation should hold if both instruments are correct?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an alternating voltage changes continuously through each cycle · aC waveforms: frequency, period and phase — taken as read

VERDICT 88W

An alternating voltage does not have one value, so any single number is a summary and the summary has to be named. The peak is the highest the waveform reaches. The RMS value is the steady voltage that would deliver the same power into a resistance. For a sine wave it is the peak divided by root two. Divide 325 by 1.414 and the meter's 230 comes straight out. This is why mains voltages are quoted as RMS everywhere: it is the figure that predicts heating and power.

TAKEAWAY 15W

Peak and RMS are different quantities, and one is the other divided by root two.

6.4 What a node has to add up to

SYSTEM OPERATIONS · A ROOM · CALLBACK · CHOICE

SCENE 31W

The busbar meter reads 900 A out. Reyes wants to know what that single figure rules out about the two circuits underneath it. A printed network diagram lies on the desk.

GUIDE 63W

All four options say something about two parallel circuits. They differ in whether they need one equation or two. Ask of each whether the meter reading alone could establish it. Charge does not pile up at a junction, so the branch currents have to add to the total. That fixes their sum and nothing else. Two of these options confuse parallel with series.

ASK 12W

What does the 900 A total tell you about the 2 circuits?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

charge does not accumulate at a junction · ohm's law and resistive networks — taken as read · charge, current, voltage and resistance — taken as read

VERDICT 85W

Charge does not build up at a junction, so current entering a node must equal current leaving it. Kirchhoff's current law gives that conservation rule. It fixes the sum of the branch currents without telling you how the total divides. The split needs another relation, such as the branch impedances. Assuming an even split is wrong when the paths differ. Assuming each branch carries the full total confuses parallel and series circuits. One equation constrains two unknown branch currents; it does not determine both alone.

TAKEAWAY 14W

A conservation law bounds the parts of a system without identifying any of them.

END OF THE DAY 17W

A reading is a claim about one quantity, and two instruments can disagree while both are right.

Where the current actually goes

PLAN CARD 111W

Tuesday, a week on, and the corridor has two circuits in it rather than one. Piotr Novak points out that this does not halve anything. The flow divides according to what each path costs it, and the newer circuit is the lower-impedance one, so it is taking two thirds of the load while the older one idles. Thabo Dube is waiting to be told which one to switch out for the morning's work. Today you work out how the current splits, decide what a node can and cannot hide, and say which arrangement leaves the least exposure. Switch the wrong one and the remaining circuit finds out what the sum was.

BRIEFING 13W

Two circuits share the same corridor and they are not sharing it equally.

WHAT YOU DECIDE 11W

Work out how flow divides between paths before switching changes them.

7.1 How the flow splits between two paths

TRANSMISSION & PROTECTION · A ROOM · BALLPARK

SCENE 34W

900 amps leave the busbar for the same destination by 2 routes. The older circuit is six ohms, the newer one three, and only the total is metered. The control-room phone is already ringing.

GUIDE 60W

Four numbers, and one of them is the busbar voltage, which cancels out of a divider. Both circuits connect the same two busbars, so they see the same voltage. Ask of each number whether it describes this path, the other path, or both. Current divides against impedance, so the path being asked about is not the impedance the formula wants.

ASK 9W

Estimate the current flowing in the older, higher-impedance circuit.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two paths between the same two points carry the same voltage across them

VERDICT 81W

Both circuits connect the same two busbars, so they have the same voltage across them. Current therefore divides inversely with impedance. For two parallel paths, one path takes the other path's impedance divided by the sum. With $6\ \Omega$ and $3\ \Omega$, the lower-impedance circuit carries twice the current of the older 1. This is why a new parallel circuit does not split flow evenly. The total busbar current constrains the sum, but the impedances determine the share in each route.

TAKEAWAY 10W

The lower-impedance path takes the larger share, in inverse proportion.

7.2 What is worth knowing before the peak

SYSTEM OPERATIONS · A PERSON · VALUE

SCENE 35W

Reyes has a small budget and four things she could find out before the evening peak. Only some of them would change what she does about the reserve. The shift log is open beside them.

GUIDE 57W

Five evidence credits, four things you could find out, and a reserve decision at six o'clock. Open each card and ask what its result would change. A measurement that confirms something already settled is not worth a credit, however clean it would be. Buy the ones that could still move the decision, and commit before the peak.

ASK 7W

Which measurements are worth buying before 18:00?

BEHIND THE BUTTON 145W

What makes evidence worth its credit. Two possible results that point at different actions. If both outcomes leave you doing the same thing about the reserve, the measurement is interesting and not useful — and tonight the difference between those two words is a credit you cannot get back.

Why the deadline is part of the question. A result that arrives at seven is a result about a decision already made. So a slower, better measurement can be worth less than a rough one in time, which is the opposite of how the same choice would be scored in a laboratory.

What the shift log is doing open beside her. It records what has already been decided and why, which is the fastest way to see which of the four questions are still open. Half of buying evidence well is knowing what is already settled.

TAKES AS READ

reserve is held against uncertainty rather than only the forecast centre · storage power and storage deliverability are separate claims · electrical power and energy over time — taken as read

VERDICT 86W

Reserve depends on uncertainty in demand and on resources that can actually deliver when called. Measuring this winter's cold-load sensitivity can change the forecast width before the peak. Re-testing storage at the relevant discharge rate can change how much reserve the store deserves credit for. Repeating a generator test that is already current mainly buys reassurance. A feeder survey that arrives in six weeks may be useful planning evidence, but it cannot change tonight. Evidence has value only when its result can still alter the decision.

TAKEAWAY 18W

The value of a measurement is the decision it can change, not how precise or interesting it sounds.

7.3 Why the far end sits low

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · BALLPARK

SCENE 34W

The substation end of the feeder is at nominal voltage. Customers four kilometres along it are complaining, and the only meter is back at the substation. The room is waiting for a written instruction.

GUIDE 61W

Six numbers, and one of them is the feeder's nominal voltage, which the drop does not use. Two pairs go together: resistance with the in-phase part, reactance with the out-of-phase part. Ask of each number which half of the impedance it belongs to. And note the rating. Eleven kilovolts is measured between lines, which is why a factor arrives with it.

ASK 11W

Estimate the volt drop from the substation to the far end.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current flowing through a resistance produces a drop in voltage

VERDICT 81W

A real feeder has both resistance and reactance. Current through those terms produces a voltage drop whose size depends on the load phase angle. The resistive part follows the in-phase current. The reactive part follows the out-of-phase current. For a balanced three-phase feeder expressed with line-to-line voltage, the result carries a factor of root three. That factor matters here because 11 kV is a line-to-line rating. Reactive support can raise the far-end voltage by reducing the reactive part of the drop.

TAKEAWAY 15W

Volt drop is set by the current and by both parts of the line's impedance.

7.4 How much power actually left the system

GENERATION • GENERATION HALL • A ROOM • CALLBACK • BALLPARK

SCENE 33W

Haddad, the generation lead, needs a number before starting anything. The only clean evidence is the first two seconds of the trend and the stored-energy figure for the machines that were online overnight.

GUIDE 66W

Five numbers, and two of them belong to the aftermath rather than the first instant. The settled deviation is what the governors did afterwards. The generation online is the denominator of a different question. Ask of each number whether it describes the first two seconds. A figure that is right about a later moment is wrong here, and the control room acts on whatever comes out.

ASK 10W

Estimate the generation lost at the moment of the trip.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate of change can be read off the steep part of a trend • synchronous machines and synchronisation — taken as read • electrical power and energy over time — taken as read

VERDICT 71W

Immediately after a generator trip, governor and reserve actions have not yet changed the power balance. Stored kinetic energy supplies the shortfall for the first moments, so frequency moves. With stored kinetic energy E_k , nominal frequency f_0 and measured rate df/dt , the imbalance is $\Delta P = 2E_k |df/dt| / f_0$. Using 25,000 MW·s, 50 Hz and 0.30 Hz/s gives about 300 MW. This estimate is most useful before slower controls begin reshaping the trace.

TAKEAWAY 15W

The initial rate of frequency change measures the size of the imbalance, given the inertia.

END OF THE DAY 14W

Current divides between parallel paths in inverse proportion to their impedance, whatever anybody intended.

80 milliseconds

PLAN CARD 106W

Wednesday, and a cable failed inside the substation at 13:40. The protection cleared it in under a tenth of a second, which is the whole story: nobody in the control room saw the fault, they saw the aftermath. Piotr Novak needs the fault current worked out before he will sign the switchgear back into service. June Farrow wants to know how a current transformer produces a reading from a wire it never touches. Ewa Kowalczyk has a crew standing at a gate and a rule about earthing that they are quietly impatient with. Today you size the fault, explain the instrument, and decide who may approach.

BRIEFING 17W

A cable failed at the substation and the protection cleared it before the alarm reached the room.

WHAT YOU DECIDE 11W

Establish what a fault does before anybody can respond to it.

8.1 What a short circuit is limited by

TRANSMISSION & PROTECTION · A ROOM · BALLPARK

SCENE 31W

The cable failed phase to earth. Phase voltage is 6,350 volts and the total impedance back to the source is 0.42 ohms. Novak wants the current the switchgear had to interrupt.

GUIDE 59W

Four numbers, and two of them belong to the healthy circuit and the protection. Normal load current is set by equipment drawing power, and a fault bypasses that equipment. Ask of each number whether it describes the load or the path back to the source. The clearing time decides how long the current flows, not how large it is.

ASK 8W

Estimate the current that flowed during the fault.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a short circuit leaves almost no impedance in the path

VERDICT 81W

Normal load current is set by the equipment drawing power. A fault bypasses that load, so the current is instead limited mainly by the source and network impedance back to the fault. Using the phase voltage and the Thevenin impedance gives the first symmetrical RMS estimate. Here 6,350 V divided by 0.42 Ω is about 15 kA. That is why switchgear has a separate interrupting rating. Clearing time then controls how long the large current can deposit heat and mechanical stress.

TAKEAWAY 17W

Fault current is the driving voltage over the impedance of the path, and nothing else limits it.

8.2 A reading taken without a connection

METERING & STANDARDS · A PERSON · CONTROL

SCENE 35W

June Farrow, the metering engineer, has a protected low-voltage current-transformer test loop beside the yard diagram. The bench shows primary current, secondary current and a movable core. She wants a cause, not a remembered definition.

GUIDE 66W

The rig is a current transformer: a conductor carrying current, and a core around it with a winding that reports a scaled copy. The number you are watching is that secondary current. Three controls can be changed, and a response counts only if it is larger than the bench's own wander. Change one, run it, put it back, and name the control the secondary current follows.

ASK 17W

Which change makes the secondary current disappear, and does the signal return when that change is reversed?

BEHIND THE BUTTON 173W

How a current transformer works, and why nothing touches the conductor. Current in the primary makes a changing magnetic field, the core carries that field around the winding, and the changing field induces a proportional current in it. Nothing is connected to the primary at all — which is exactly why the instrument is safe on a live circuit, and why what it needs is change rather than contact.

Why a steady current reads nothing. Induction responds to a field that is changing. Hold the primary current constant and the field stops changing, so the winding reports nothing while the conductor is still carrying full load. An instrument that reads zero on a live conductor looks broken and is behaving exactly as designed.

Why the core is the third control. Open the magnetic path and the field no longer links the winding, so the secondary falls away even with a changing primary. Two different controls therefore kill the reading for two different reasons, and telling them apart is what the reversal is for.

TAKES AS READ

a changing magnetic field can induce voltage in a nearby coil · the bench current transformer has 1 primary turn and 20 secondary turns · aC waveforms: frequency, period and phase — taken as read · charge, current, voltage and resistance — taken as read

VERDICT 82W

A current transformer needs no electrical connection between primary and secondary. Alternating primary current creates changing magnetic flux in the core. Faraday's law then induces an emf in the secondary, and the turns ratio sets the secondary current approximately. A steady DC current can still create a magnetic field. After the switching transient, however, the flux stops changing and the

secondary signal falls almost to zero. Removing the suspected cause and then restoring it is stronger evidence than changing several controls together.

TAKEAWAY 17W

A current transformer responds to changing magnetic flux; a steady primary current produces no sustained secondary signal.

8.3 The peak, and how far the cold moves it

LOAD & FORECASTING · A ROOM · BALLPARK

SCENE 36W

The base peak for a Friday in this week of the year is 6,400 megawatts. Lindgren's own figure for this winter is 120 megawatts of extra demand per degree below 0, and tomorrow is minus 4.

GUIDE 63W

Four numbers, and one of them is the reserve held, which belongs to the width of the forecast rather than its centre. Ask of each whether it is a base, a sensitivity, or a temperature. A sensitivity is megawatts per degree on this system, measured from this winter. What comes out is the centre of a forecast, and the reserve covers the rest.

ASK 4W

Estimate tomorrow's evening peak.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

demand rises as the temperature falls, roughly in step · electrical power and energy over time — taken as read

VERDICT 79W

A load curve is the same shape most days, because people do the same things at the same times. What moves it is weather, and the useful form of that is a sensitivity: how many megawatts one degree is worth on this system. Measured from this winter's own data rather than from a textbook. Base plus sensitivity times degrees gets the centre of the forecast. It does not get the width, which is what the reserve has to cover.

TAKEAWAY 17W

A forecast is a base shape plus a sensitivity, and the sensitivity is measured rather than assumed.

8.4 Which of the three cannot wait

SYSTEM OPERATIONS • A ROOM • CALLBACK • DELEGATE

SCENE 35W

Reyes has three live problems on the board and only one attention channel she can keep herself. Aaron Whitlock is ready to take a handoff, but she refuses any instruction that amounts to “watch it.”.

GUIDE 60W

Three live problems and one attention channel you can keep. For each of the other two, name an owner, the first real action they will take, and the reading or event that brings the problem back to you. Reyes will not accept anything that amounts to 'watch it'. Then take the watch on the one that needs your own judgement.

ASK 11W

What do you keep, and what exactly do you hand off?

BEHIND THE BUTTON 138W

Why 'watch it' fails. An owner who has been told to observe will come back at the moment they need instructions, which is the moment you are busiest. A first action is what lets somebody hold a problem without you, and naming it is the difference between delegating and deferring.

What a return condition is for. It is not a promise to check in. It is the threshold at which the problem stops being theirs, agreed in advance, so nobody has to judge in the moment whether this is bad enough to interrupt you.

How to choose what to keep. Compare the three on how fast they are getting worse, on whether they are already contained, and on what happens if nobody acts. The one that needs judgement rather than procedure is the one you cannot hand over.

TAKES AS READ

a stable problem can be severe without becoming more urgent each minute • reserve actions need lead time before the peak

VERDICT 86W

The cable outage is loud but stable: customers are already off and a crew is moving. The fuel-limited machine has a known time and quantity, so another operator can prepare its replacement. The forecast error is different. It is still growing, and reserve procurement needs lead time before the evening peak. Every update that passes without action removes options that cannot be recovered later. Delegation is therefore not “watch it.” Each handoff needs an owner, a first action and a condition that brings the problem back.

TAKEAWAY 22W

Urgency comes from how fast a condition is worsening, how reversible it is, and whether somebody else can own the next move.

A fault is limited only by impedance, which is why the numbers are enormous and the times are tiny.

The system splits

BEFORE THE DAY — CANVASS

Whose lights are actually back

The system says eleven feeders are restored. The complaint line says otherwise, and the difference matters because a feeder that reads energised at the station and dead at the street is a fault somewhere in between. Ask along the affected streets until enough answers have come back to tell those two possibilities apart.

PLAN CARD 134W

Friday, 18:20, and the corridor tripped at the worst hour of the worst day — the same corridor the room chose to hold inside its emergency rating a week ago, on a thermal record nobody had checked. The valley did not go dark. It separated, and it is now an island of its own with the ridge, one gas machine and about a fifth of the demand it had at lunchtime. Dolores Reyes has minutes to decide whether to help it survive or shut it down deliberately. Nadia Haddad says the one machine in there has almost no inertia behind it. Chinelo Obi has the customer numbers. Today you work out what is holding that island's frequency up, decide what stored energy is actually available, and choose whether to keep it or collapse it.

BRIEFING 15W

The corridor tripped and the valley is now running on its own, for the moment.

WHAT YOU DECIDE 15W

Work out what an island is, and whether the one that has formed can hold.

9.1 Why the island's frequency is falling slowly

SYSTEM OPERATIONS · A ROOM · DEGENERACY

SCENE 36W

The island frequency is falling at 0.10 hertz per second. Its dispatchable machine is already at full output, while inverter-connected ridge generation is low. Haddad wants the power shortfall, but the wall gives only the slope.

GUIDE 65W

Two controls slide along a family of combinations that all reproduce the observed 0.10 hertz per second. Find out how wide that family is before naming a number. The wall gives only the slope, and a slope alone cannot separate a large shortfall on a heavy system from a small one on a light system. Then bring in the stored kinetic energy, which collapses it.

ASK 18W

What power shortfall is consistent with the 0.10 Hz/s fall once the island's stored kinetic energy is measured?

BEHIND THE BUTTON 136W

Why the slope alone is not enough. Frequency falls at a rate set by the power shortfall divided by the stored kinetic energy of everything spinning. One equation, two unknowns. Every pair that gives the same ratio produces exactly the same slope on the wall.

Why the ridge generation matters. Inverter-connected machines contribute little or no stored kinetic energy — they are not spinning masses locked to the frequency. So an island leaning on them has less inertia than its megawatts suggest, and the same shortfall drives the frequency down faster.

What measuring the inertia buys. It fixes one of the two unknowns, so the shortfall follows immediately. That is why system operators now measure inertia rather than assuming it: the assumption was safe while the fleet was synchronous and stopped being safe as it changed.

TAKES AS READ

the island is near 50 Hz · the initial frequency slope follows $df/dt = f_0 \Delta P / (2E_k)$

VERDICT 81W

The initial frequency slope comes from two quantities at once: the real-power mismatch and the stored kinetic energy resisting it. A slow fall can therefore mean a small shortfall, large inertia, or a matching combination of both. Inverter-connected wind can add real power without adding the same synchronous kinetic energy, so installed megawatts do not settle the question. First trace the combinations that reproduce 0.10 Hz/s. Then use an independent estimate of stored kinetic energy to select the matching power shortfall.

TAKEAWAY 21W

A frequency slope constrains the ratio of power imbalance to stored kinetic energy; it does not determine either one by itself.

9.2 What is stored, and for how long

METERING & STANDARDS • A PERSON • PROTOCOL

SCENE 30W

Farrow is asked to list what the island has to draw on in the first ten seconds. Two of the four things on her list cannot help at that timescale.

GUIDE 58W

Four stores of energy, and the right column is four timescales. Pair them by asking how soon each one answers, and for how long. Two of these are on the list and cannot help in ten seconds. The fields in transformers and lines hold real energy for a fraction of a cycle, which is nothing you can dispatch.

ASK 9W

Say where an island's short-term energy actually comes from

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

energy can be stored in a spinning mass, in a magnetic field or in a chemical cell • aC waveforms: frequency, period and phase — taken as read • charge, current, voltage and resistance — taken as read

VERDICT 81W

Each store answers over a different interval and confusing them is how a plan fails. Rotational energy in a spinning machine is there instantly, without anybody asking, and it is gone in seconds. A battery through an inverter responds in well under a second and can hold for hours. That makes it the most useful thing an island can have. Energy in the magnetic fields of transformers and lines is real, and it is a fraction of a cycle: nothing dispatchable.

TAKEAWAY 16W

Every store has a timescale, and matching the store to the timescale is the whole trick.

9.3 Keeping it, or collapsing it on purpose

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CHOICE

SCENE 37W

Obi has 26,000 households inside the island and a frequency still falling. Holding it needs demand off. Collapsing it means all of them, and a longer way back. The event board is still showing the overnight state.

GUIDE 60W

Four options, and the island has one dispatchable machine already at maximum. So ask what lever each option actually uses. Two of them are bets on something outside the room. One costs all twenty-six thousand households and a restoration measured in hours, because a dead network is built back piece by piece. The frequency is still falling while you choose.

ASK 7W

What should be done with the island?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

restoring a collapsed island takes longer than holding a live one

VERDICT 78W

An island with its only dispatchable machine at maximum has one lever left, and that is demand. Shedding a fifth stops the fall and keeps four fifths of the valley supplied, which is a far better outcome than everybody off. Letting it collapse costs all 26,000 households and buys a restoration measured in hours. That is because a dead network has to be built back piece by piece. Waiting on wind is betting the island on a forecast.

TAKEAWAY 16W

Shedding some load to keep an island alive is usually cheaper than restoring it from nothing.

9.4 Loss when the load turns up anyway

TRANSMISSION & PROTECTION · A ROOM · CALLBACK · BALLPARK

SCENE 36W

With the feeder out, flow has redistributed and the corridor is carrying more. Novak wants the heat figure before he decides whether the evening needs a switching change. A printed network diagram lies on the desk.

GUIDE 66W

Five numbers, and two of them are the current an hour ago and the conductor temperature. Neither belongs in the loss. Ask of each number whether the heat depends on it. And note the square. Current has risen about eleven per cent, and the heat rises by nearly a quarter. The rating is thermal, so that is a bigger bite of the margin than it looks.

ASK 10W

Estimate the power now lost as heat in the corridor.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

loss in a conductor rises with the square of the current · ohm's law and resistive networks — taken as read · electrical power and energy over time — taken as read

VERDICT 72W

The same loss relation now gives a worse answer. Resistive heating rises with the square of RMS current, across all three phases. Current has risen by about 11%, from 1,150 A to 1,280 A. The heat rises by about 24%, from roughly 17 MW to 21 MW. That matters because the conductor rating is thermal. A small-looking increase in current can therefore use a much larger fraction of the available temperature margin.

TAKEAWAY 17W

When a quantity enters squared, a small change in it is a large change in the result.

END OF THE DAY 16W

An island lives or dies on whether generation inside it can follow the load inside it.

Cranking path

PLAN CARD 111W

Saturday, 19:04, and the island went. The valley is dark and every machine in it needs electricity to start: the boiler feed pumps, the fans, the control systems, the switchgear itself. Nadia Haddad has one small hydro set at the head of the valley that can start on its own, and Thabo Dube has a yard full of open switches. Dolores Reyes has to decide the order, and there is only one that works. Today you cost the energy the valley has already lost, work out why the small set matters more than the large one, and write the restoration in the sequence physics allows rather than the one people want.

BRIEFING 17W

The valley collapsed at 19:04 and nothing in it can start without power it has not got.

WHAT YOU DECIDE 15W

Build a live system out of a dead one, in the only order that works.

10.1 What the dark hours actually cost

GENERATION · GENERATION HALL · A ROOM · BALLPARK

SCENE 33W

The valley was drawing about 300 megawatts when it went. Haddad expects the restoration to take two and a half hours, and the report will want the figure in energy rather than power.

GUIDE 61W

Four numbers, and two of them belong to other questions: the household count and the frequency before the split. Ask of each whether it is a power, a duration, or neither. Power says how fast energy is delivered. Energy says how much arrived, and it is what a settlement runs on. Both get called demand, which is where this goes wrong.

ASK 8W

Estimate the energy not delivered during the restoration.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

power sustained over a period of time delivers energy · charge, current, voltage and resistance — taken as read

VERDICT 84W

Power says how fast energy is being delivered. Energy says how much arrived. The two are confused constantly, because both get called demand. Multiply the power by the time it was sustained and the result is energy. Megawatt-hours, if the power was in megawatts and the time in hours. That is the number every settlement, every compensation claim and every inquiry runs on, because it is the one that scales with duration. Two systems can lose the same megawatts and owe entirely different amounts.

TAKEAWAY 16W

Power is a rate and energy is the amount, and the settlement is always in energy.

10.2 The order physics allows

SYSTEM OPERATIONS · A PERSON · CHAIN

SCENE 32W

One hydro set that can start unaided, a dead network, and a room full of people who want the big station back first. Reyes needs the sequence written before anybody switches anything.

GUIDE 64W

Build the path in the order the energy actually travels, from the machine that can start unaided to the one that cannot. Then name the required link that limits it. The biggest station is not the answer, and neither is the largest number on the board: a path is limited by the link with the least to give, and the readings tell you which.

ASK 15W

How does startup power reach the large station, and which required link limits the path?

BEHIND THE BUTTON 139W

What black start means. Almost every generator needs power to start: excitation, auxiliaries, pumps, controls. A hydro set with a battery, a governor and a head of water behind it does not, which makes it the only place a dead network can begin. Everything else is downstream of it.

Why the order is not a preference. Energising a path means charging line capacitance and picking up auxiliaries in an order the equipment can survive. Take too much at once and the small starting machine collapses, and the network is dead again with less to try next.

Why the room wants the wrong thing. The big station is what everybody is waiting for and what the public will notice. That is a reason to write the sequence down before anybody switches anything, rather than a reason to put it first.

TAKES AS READ

the large station needs electrical auxiliaries before it can start · the hydro unit can start without an outside electrical supply

VERDICT 87W

A dead network cannot start from its biggest generator if that machine needs pumps, fans and controls before it can turn. The black-start hydro unit works because water can supply torque without outside electrical power. Its generator then energises a bus and a narrow startup feeder to the large station. The feeder only needs to carry the station auxiliaries, not the future 500 MW output. That modest link governs the restart. Once the auxiliaries are live, the large machine can start and the system can grow outward.

TAKEAWAY 24W

A black start is a dependency chain: the largest generator is useless until a smaller source can energise its auxiliaries through a complete path.

10.3 Why it cannot go faster

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CASEBOOK

SCENE 35W

Obi has crews asking why restoration is moving a few streets at a time. The substation is live, and the whole feeder could be closed in one movement. The switching sheet is open beside him.

GUIDE 57W

Four restoration steps, and four different limits. Pair them by asking what is small: the live system, the load on the line, or the agreement between two systems. One of these is the opposite of most people's instinct. A long circuit with almost nothing on it raises the voltage at its far end rather than lowering it.

ASK 10W

Match each restoration step to the reason it is limited

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a small live system is disturbed more by a given block of load than a large one

VERDICT 89W

The limits change as the system grows. Very early, the live island is tiny and a block of load is enormous relative to it. Blocks stay small: this is a frequency limit.

Energising long circuits with almost no load on them raises voltage rather than lowering it. That is a reactive limit and the opposite of most people's instinct. Cold load picks up higher than normal, because every thermostat in the street is calling at once. And synchronising two live systems is a matching problem, not a capacity one.

TAKEAWAY 16W

Every restoration step is limited by something different, and hurrying the wrong one collapses it again.

10.4 The order the records claim things happened in

METERING & STANDARDS · A ROOM · CALLBACK · TRACE

SCENE 31W

June Farrow, the metering engineer, has three records of the same eight seconds. They disagree about what happened first. One record was stamped by a clock nobody has checked since spring.

GUIDE 56W

Open each record to see which clock and which physical input it depends on. Keep the ones that would still stand if one clock were wrong, and untick the rest. Then name the shared source behind the false early timestamps. Three records disagreeing about the order is evidence about the records, not only about the event.

ASK 16W

Which records still support the event order, and what shared source explains the false early timestamps?

BEHIND THE BUTTON 141W

Why a clock is part of the measurement. A record's timestamp is as good as the clock that made it, and a clock nobody has checked since spring can be seconds out. Every event stamped by it moves together, which is exactly what makes a false ordering look consistent.

How to tell dependence from agreement. Two records agreeing prove nothing if both were stamped by the same clock. What matters is whether a record's timing came from somewhere else — a synchronised source, a physical input with its own time base — because only those constrain the order independently.

What the order is for. Sequence-of-events analysis is how a blackout is explained: what tripped first decides whether this was a protection failure, an operator action or a cascade. Getting the order wrong produces a confident and wrong account of the night.

TAKES AS READ

a timestamp combines a measured event with a clock reference · two channels can agree because they share the same source · faraday's law and electromagnetic induction — taken as read · RMS against peak and average — taken as read

VERDICT 80W

A timestamp contains a physical event and a clock claim. Those can fail separately. Two relay-derived records can agree because they share one internal clock, so their agreement is not two independent measurements. The GPS disturbance recorder uses a different time source and preserves the physical ordering. The slower SCADA record can still confirm breaker state, even though its two-second scan cannot resolve sub-second order. Dependency tracing keeps useful evidence instead of discarding every channel touched by the bad clock.

TAKEAWAY 14W

Agreement is independent evidence only when the channels do not share the same dependency.

END OF THE DAY 18W

A dead system is rebuilt from one machine that needs nothing, outward along a path chosen in advance.

6 hours, not 40 minutes

PLAN CARD 121W

Sunday, and June Farrow has finished tracing the corridor's thermal history. The emergency rating that bought the room 40 minutes a week ago was calculated from a sensor that had been reading 11 degrees low since a firmware update in the spring. The corridor was not 40 minutes from its limit. It had been above continuous rating for six hours, and it had been there twice before. Dolores Reyes has to decide what that does to every decision taken since. Today you work out what a drifted sensor does to a rating, decide which of the week's conclusions still stand, and say what you would have done differently. Nothing in the week was reasoned badly. It was reasoned from a number.

BRIEFING 16W

Farrow has traced the corridor's thermal record, and the number the room worked from was wrong.

WHAT YOU DECIDE 9W

Re-read the week from a record nobody had checked.

11.1 The sensor that was confident and wrong

METERING & STANDARDS · A ROOM · CHOICE

SCENE 31W

Farrow has three sources for the corridor's temperature that week: the sensor, an infrared survey from the ground. The sag measured off a photograph. Two of them agree with each other.

GUIDE 55W

Four options, and three temperature sources have to be accounted for. Two agree with each other and neither of them shares the corridor sensor. Ask of each option how many of the three it explains. Small scatter shows an instrument repeats, not that it is right. And the firmware date marks where the disagreement starts.

ASK 8W

Which explanation best fits all three temperature sources?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an instrument can be precise and still be offset from the truth

VERDICT 86W

Small scatter only shows that an instrument repeats. A fixed offset can produce readings that are precise and wrong at the same time. Here 2 methods that do not share the corridor sensor both place the conductor about 11 degrees hotter. The firmware date also marks when the disagreement begins. Sag depends on tension as well as temperature, so it is not proof alone. Its value is that it corroborates an infrared method through different physics, which is exactly what a sensor offset cannot explain away.

TAKEAWAY 17W

Precision is repeatability; accuracy is being right. A drifted sensor has the first and not the second.

11.2 What a rating is a statement about

TRANSMISSION & PROTECTION · A PERSON · SWEEP

SCENE 53W

The corridor's conductors may run continuously at 75 degrees and no hotter. Above that the aluminium starts to lose strength, and the span sags towards the road beneath it. Novak has the corrected figures and wants the current that limit allows, because a rating quoted in amps is really a statement about temperature.

GUIDE 67W

Drag the corridor current. The readout is the conductor's own temperature, not its rating, and only the currents you stop at get plotted. You want the current at which that temperature reaches the 75-degree limit named above. Heating rises faster than the current causing it, so the curve steepens as you go and the last hundred amps cost more than the first. Mark that current and commit.

ASK 20W

75 degrees is the limit the conductor was built to. What current does that allow it to carry all day?

BEHIND THE BUTTON 196W

What sets the temperature. Current heats a conductor as the square of itself, and the conductor loses that heat to the air by convection and to the sky by radiation. The temperature settles where the two balance, which is why the curve is not a straight line: doubling the current asks the cooling to remove four times as much heat.

Why a rating is a temperature. The ampere figure printed on a corridor is the current that reaches the temperature limit under an assumed ambient, an assumed wind and an assumed sun. Change any of those and the same amps arrive at a different temperature. That is why a corrected weather figure can move a rating without anybody touching the line, and why a rating quoted without its assumptions is not a limit at all.

What happens above the limit. Aluminium held above its continuous temperature begins to anneal, losing strength it does not get back, and the hotter span stretches and sags. The clearance over the road below is a legal minimum, so the temperature limit is doing two jobs at once: protecting the conductor, and keeping the line where the survey says it is.

TAKES AS READ

a conductor's rating is a limit on temperature rather than on current · resistive heating rises faster than the current that causes it

VERDICT 88W

A rating is not a fuse setting. It is a temperature translated into a current, and the translation runs through the heating. That goes as the square of the current: the last hundred amps buy far more degrees than the first hundred. The curve steepens and the rating sits where it crosses the agreed temperature. Two consequences follow. A

rating is only as good as the temperature measurement behind it. That is why a drifted sensor moves a limit that looks like a fixed property of the metal.

TAKEAWAY 18W

A rating is a temperature expressed as a current, so a wrong temperature makes the current limit meaningless.

11.3 What survives a bad number

SYSTEM OPERATIONS · A ROOM · CASEBOOK

SCENE 36W

Reyes has the week's decisions on the board and one wrong input underneath several of them. She wants each one marked as still good, now unsupported, or wrong. The yard radio is still live beside them.

GUIDE 59W

Four decisions from the week, and one wrong input underneath some of them. Trace each decision back to what it was argued from. Ask whether temperature is in that chain at all. A loss calculation that used current and resistance never touched the sensor. Neither did a frequency. One wrong number voids what leaned on it and nothing more.

ASK 8W

Decide which of the week's conclusions still stand

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a conclusion can depend on one input or on several

VERDICT 85W

One wrong number does not void a week. It voids whatever leaned on it, and the job is tracing which. The choice to hold the corridor rested entirely on the duration, so it falls. The loss calculation used current and resistance and never touched temperature, so it stands. The restoration order was argued from consequence and household counts, which the sensor never entered. And the decision to shed a fifth of the island was taken on a frequency the sensor has nothing to do with.

TAKEAWAY 20W

A bad input invalidates what rested on it and leaves the rest standing, and telling them apart is the work.

11.4 What can come off, and what cannot

LOAD & FORECASTING · A ROOM · BELT

SCENE 34W

A shed order needs a list in ten minutes. Some feeders can be dropped and restored an hour later with nobody harmed. Others carry something that cannot simply be switched off and switched back.

GUIDE 57W

Two bins. A feeder can be shed if the load behind it tolerates an hour without power and comes back the way it went off. It cannot be shed if something behind it needs power to stay safe, or cannot be restarted at will. Sort on what is behind the feeder, not on how important it sounds.

ASK 15W

Send each feeder to the bin that says whether the shed order can take it.

BEHIND THE BUTTON 97W

Why sheddable is not the same as unimportant. A large industrial customer with its own standby plant is sheddable; a small clinic on one supply is not. Size, revenue and prominence have nothing to do with it, which is why the list is written in advance and not argued out at the time.

What makes a load unsheddable. Something behind it that fails unsafely without power, or that cannot be brought back on demand — a furnace that sets solid, a pumped drain that floods, a process that has to be run to a finish once started.

TAKES AS READ

some loads can be interrupted and restored, and some cannot

VERDICT 141W

A shed order is written to the state of what is behind each feeder. On one side sit the loads that tolerate an hour off and come back exactly as they went — cold stores with thermal mass, offices, street lighting, a works with its own standby set. On the other side are loads where an hour without power is not an hour of inconvenience: a hospital on one supply, a smelter whose bath sets solid, a pumped drain under a town, a process that has to run to a finish once it has started. None of that maps onto size or importance, which is exactly why the list is prepared cold and not decided at four in the morning by whoever is loudest on the phone. The one thing a dispatcher cannot do is work it out during the event.

TAKEAWAY 18W

What can be shed is decided by what is behind the feeder, not by how important it sounds.

11.5 What is common and what divides

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CALLBACK · PROTOCOL

SCENE 30W

Obi, the distribution supervisor, is training a new starter. Four circuit sketches are on the whiteboard, and he wants each matched to the quantity that stays common across that arrangement.

GUIDE 61W

Four arrangements, and the right column names what stays common in each. Pair them by asking how many paths the current has. One path means one current and a divided voltage. Two nodes shared means one voltage and a divided current. Then ask what adding a branch does to the total. Getting this wrong sizes a fuse for the wrong current.

ASK 7W

Match each connection to what it shares

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

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TAKES AS READ

a circuit can be connected end to end or side by side · kirchhoff's laws and nodal analysis — taken as read

VERDICT 86W

Series and parallel circuits differ by what stays common. In series there is one path, so every element carries the same current. The supply voltage divides across the elements. In parallel every branch connects to the same two nodes, so each branch has the same voltage. The current divides instead. Adding a parallel branch lowers the equivalent resistance and raises total current. Adding a series resistance raises the equivalent resistance and lowers the loop current. Mixing these rules leads directly to wrong fuse and conductor sizing.

TAKEAWAY 16W

Series shares the current and divides the voltage; parallel shares the voltage and divides the current.

END OF THE DAY 23W

A conclusion is only as good as the measurement under it, and a drifted instrument leaves the reasoning intact and the answer wrong.

A shift where nothing happens

BEFORE THE DAY — EVADE

The contractor who wants a decision in the yard

A contractor has been in the yard since six wanting a verbal instruction about the metering hut, and Reyes, the shift supervisor, has said plainly that nothing gets agreed on the ground without the paperwork. Keep out of his way and get your own round done. Anything agreed in a yard at seven in the morning is agreed by nobody in particular.

PLAN CARD 120W

Monday, and it is a quiet shift. The forecast is holding to within 30 megawatts, every corridor is comfortably inside rating, the ridge is producing steadily, and there is nothing on the board that needs a decision. Aaron Whitlock has the log open with almost nothing in it. Dolores Reyes uses these evenings to walk the room and ask what everybody is assuming, because a quiet shift is the only time anybody has to. Today you check what the system would do if one thing went, read a trend that is behaving, and say which of tonight's comforts is actually a measurement and which is a habit. Nothing will go wrong tonight. Nothing went wrong the Monday before last, either.

BRIEFING 12W

The forecast holds, every circuit is inside limits, and nothing needs deciding.

WHAT YOU DECIDE 12W

Run an ordinary evening, and notice what ordinary is now resting on.

12.1 The failure that has not happened

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 38W

The screen says every single contingency is survivable tonight. Whitlock asks what the check does not cover, and Reyes says that is the right question on a quiet shift. The event board is still showing the overnight state.

GUIDE 58W

All four options describe something a clean contingency check might mean. They differ in scope: one failure or two, survivable or likely. Ask of each whether the check's own list could support it. The check is exactly as broad as that list. Last Friday was a corridor already above a limit nobody knew about, and then a trip.

ASK 9W

What does a clean contingency check tell you tonight?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a system is required to survive any one credible failure

VERDICT 120W

The check is exactly as broad as its list. It takes each credible single failure in turn, works out the flows afterwards, and confirms nothing ends up over a limit. That is a genuinely strong statement and a narrow one. It does not cover two failures at once, it does not cover a failure outside the list. It says nothing at all about what is likely — only about what is survivable. Last Friday was two things: a corridor already above a limit nobody knew about, and then the trip. When a second thing lands on neighbours checked only one at a time, each trips correctly in turn, and the sequence of events is what has to be reconstructed afterwards.

TAKEAWAY 15W

An N-1 check is a statement about single failures, and it says nothing about 2.

12.2 The trend with nothing wrong with it

LOAD & FORECASTING · A PERSON · PROTOCOL

SCENE 34W

Lindgren's forecast has tracked within 30 megawatts all evening. He wants to know what that does and does not license the room to assume about tomorrow. The room is waiting for a written instruction.

GUIDE 64W

Four observations about a forecast, and the right column says what each one licenses. Pair them by asking where in the distribution that day sits. A mild Monday is the middle, and the middle is where the model was best calibrated. Only one of these four is evidence about the method rather than about a night, and the room is deciding reserve on it.

ASK 6W

Read a forecast that is behaving

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a forecast can be right by luck as easily as by skill on any one day · charge, current, voltage and resistance — taken as read

VERDICT 77W

A forecast is a distribution. Landing inside 30 megawatts on a mild Monday is what the middle of that distribution looks like and it tells you almost nothing. That is because mild Mondays are the days the model was best calibrated for. The evidence that matters is how it behaves on the days it was least calibrated for, which are rare by construction. So a run of good nights is not a reason to hold less reserve.

TAKEAWAY 18W

One good night is not evidence about the method, because the width was never a claim about tonight.

12.3 Which of these is a measurement

GENERATION • GENERATION HALL • A ROOM • VALUE

SCENE 30W

Haddad lists four reasons the room feels comfortable tonight. Three are current measurements or long-run evidence. One is an inherited stored-energy figure written before the inverter-connected fleet became much larger.

GUIDE 52W

Four reasons the room feels comfortable, and four check credits. Open each one and ask what it actually is: a current measurement, long-run evidence, or a number somebody inherited. Buy the evidence that could change the under-frequency timing. A figure written before the fleet changed is a claim about a different system.

ASK 13W

Which evidence should be bought before the room trusts its present under-frequency timing?

BEHIND THE BUTTON 144W

Why the stored-energy figure is the suspect. Under-frequency protection is timed against how fast the frequency can fall, and that depends on the inertia of everything spinning. As inverter-connected generation grew, the inertia fell — but a number written down before that shift keeps being quoted because it has been in the file for years.

What separates a measurement from a comfort. Three of the four can be pointed at something measured recently on this system. The fourth is an inheritance, and the room's confidence rests on it without anybody having decided to trust it.

Why timing is where it bites. Under-frequency load shedding has to act before the frequency reaches the point where machines trip. If the real inertia is lower than assumed, the frequency gets there sooner and the settings are late — which is a blackout rather than a load-shedding event.

TAKES AS READ

stored kinetic energy affects the initial rate of frequency change • inverter-connected generation may add power without adding synchronous inertia

VERDICT 80W

Three comforting numbers are current measurements or long-run evidence. The inherited inertia figure is different. The fleet has changed, and inverter-connected generation can supply real power without the same synchronous stored kinetic energy. That matters because stored energy sets how fast frequency moves after a sudden imbalance. Recomputing it from current machine data can therefore change protection timing immediately. If the old 25,000 MW·s figure has fallen to 18,000 MW·s. The same 300 MW trip moves frequency about 40% faster.

TAKEAWAY 18W

An old assumption deserves priority when it is load-bearing, likely to have changed, and cheap enough to check.

12.4 What the desk is chasing now

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · SPOT

SCENE 42W

Faults arrive on the board all evening. The desk works to one standing priority, and the shift manager changes it as the picture changes — after the storm crosses, after the hospital calls, after the count of customers off passes ten thousand.

GUIDE 55W

Take the faults the standing priority currently wants and leave the rest. The priority at the top of the board is replaced during the evening and nobody announces it.

What is scored is the faults either side of a change, because they are the only ones that show whether the desk is reading the board.

ASK 12W

Dispatch to the priority on the board, and keep watching the board.

BEHIND THE BUTTON 97W

Why the priority changes at all. Early in an event the work is restoring the most customers per crew-hour. Once a hospital is on generator the work is that hospital. After midnight it is whatever cannot be left overnight in the weather. Each of those is a different order of the same list.

Why the changeover is where it goes wrong. A desk that keeps dispatching to the old priority for ten more minutes sends its last free crew to the job that mattered an hour ago, and nothing on the board says that is what happened.

TAKES AS READ

a control desk works to a stated priority that somebody can change

VERDICT 121W

Three priorities run across the evening. Each one wants a different part of the board. Most customers restored first. Then anything with a vulnerable customer behind it. Then anything that cannot be left out overnight in the weather. A fault can answer two of those at once, and that is what makes the switch cost real. What the panel scores is the window either side of each change. Everything else on the board is wanted by neither priority, so it is left alone correctly by somebody who has read nothing. The desk's own version is the ten minutes after a hospital calls. The priority has already changed, and the crews rolling are rolling to the order in force when they left.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

12.5 What you do before a relay does it for you

TRANSMISSION & PROTECTION · A ROOM · CALLBACK · TRIGGER

SCENE 34W

The corridor is at 108% of continuous rating and rising slowly. Novak, the transmission engineer, wants a response he can hand to Dube, the switching operator, before the relay makes the choice for them.

GUIDE 75W

One rule, written before the trend arrives and fixed once you release it. Moving 200 MW off the corridor takes six hours from the moment it is called, and at 120 % of continuous rating the protection takes the line out itself. So the line goes back from 120 by however far the loading climbs in those six hours — and not so far back that you are redispatching a corridor which has shown nothing.

ASK 20W

At what loading do you commit the redispatch, given the move takes six hours and protection acts at 120 %?

BEHIND THE BUTTON 140W

Why act before the relay. Protection will disconnect the corridor when its limit is reached, and it will do so without regard to what else that leaves connected. Anything you do before then is a controlled choice; anything after is a consequence.

Why two thresholds rather than one. Redispatching generation takes minutes and shedding load takes seconds. A single threshold set for the slower action fires far too early for the faster one, and set for the faster one leaves no time for the slower. The lead time is the reason there are two numbers.

What Dube needs from this. A switching operator acting at three in the morning needs a rule, not a judgement. Numbers agreed in advance are what make the night's actions defensible afterwards, and what stop the decision being made by whoever happens to be awake.

TAKES AS READ

continuous and emergency ratings are different limits · redispatch is slower than automatic or operator load shedding · charge, current, voltage and resistance — taken as read · ohm's law and resistive networks — taken as read

VERDICT 101W

An emergency rating is a time-limited thermal allowance, not permission to wait until the relay acts. Redispatch takes six hours from the moment it is called, so the loading at which it is committed has to sit back from the 120 per cent ceiling by however far the corridor climbs in those six hours. That is what makes the number smaller than the number the ceiling suggests. Writing the rule before the trend appears is what keeps hindsight from turning into policy, and a threshold that is physically sensible but fires after its action can still arrive is a failed rule.

TAKEAWAY 20W

A protection plan must be written before the number moves, with earlier thresholds for actions that take longer to arrive.

END OF THE DAY 15W

A quiet system is a set of assumptions nobody is currently being forced to check.

Merit order

PLAN CARD 112W

Thursday, and tomorrow has to be bought tonight. Sten Lindgren has a peak that moves with the temperature and a width he will not narrow. Mina Sarraf has a ridge that will produce almost nothing for money and cannot promise any of it, and she is tired of being asked to. Nadia Haddad has expensive machines that can be told exactly what to do. Dolores Reyes has to sign an order that puts them in sequence. Today you forecast the peak from the load curve, decide what merit order actually means when the cheapest plant is the least certain, and choose what to do with more wind than the corridor can carry.

BRIEFING 18W

The cheapest generation is on the ridge and it will not say what it is doing after four.

WHAT YOU DECIDE 15W

Decide what runs tomorrow, and what is held back for the day it is needed.

13.1 The ground is not at one voltage

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CHOICE

SCENE 38W

The cable is isolated at both ends and a tester reads dead at the work point. Kowalczyk, the safety lead, stops the crew because one protection step is still missing. The substation remains connected to live equipment nearby.

GUIDE 59W

All four options are things a crew could do next, and three of them sound careful. Ask of each what it would protect against: a source you know about, or one that arrives later. Isolation removes the known sources. A dead test confirms one moment. Neither holds the conductor down if switching changes or a voltage is induced nearby.

ASK 10W

What must happen after isolation and a successful dead test?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a conductor can be re-energised from more than one direction · charge, current, voltage and resistance — taken as read

VERDICT 81W

Isolation and proving dead answer different questions. Opening both ends removes the known sources, while the dead test confirms the condition at that moment. Neither guarantees the conductor will stay safe if switching changes or induced voltage appears. Portable earths provide a low-impedance path for unexpected current and keep the work conductor close to local ground potential. That is why earthing follows the dead test rather than replacing it. The sequence turns a measured safe state into a maintained safe state.

TAKEAWAY 13W

Isolation is not proof of safety, and earthing is what makes it provable.

13.2 Cheapest first, and the exceptions that are not exceptions

SYSTEM OPERATIONS · A ROOM · CASEBOOK

SCENE 30W

Four plants and an order to sign. Reyes wants each one matched to the actual reason it sits where it does in the stack, rather than to its price alone.

GUIDE 64W

Four plants and four reasons to sit where they do. Only one of the four reasons is price. Pair them by asking what bends the merit order: reliability, the ability to move quickly, or where the plant is. A machine at full output cannot sell reserve. A cheap machine in the wrong place can be worse than an expensive one in the right place.

ASK 11W

Match each plant to why it is dispatched where it is

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

generation is scheduled in order of what it costs to run

VERDICT 89W

Dispatch starts from cost, because running the cheapest first is what makes electricity affordable. Then three things bend it. Plant with almost no running cost goes first and cannot be relied on. It displaces fuel without displacing the need for something reliable behind it. Plant that can change output quickly is worth holding part-loaded, because reserve is a service and a machine at full output cannot provide it. And plant in the right place can be needed regardless of price, when the alternative is a corridor over its limit.

TAKEAWAY 15W

Merit order is cost order, corrected for what the system also needs from a machine.

13.3 When the cheapest power cannot get out

GENERATION · GENERATION HALL · A ROOM · SCIENCETANK

SCENE 37W

Sarraaf, the renewable operator, can produce 340 megawatts from the ridge tonight. The corridor out of the valley can carry about 190. There is no second export path available. A printed network diagram lies on the desk.

GUIDE 96W

The ridge can produce 340 MW tonight and the corridor is rated at about 190 MW continuous. Nights like this occur perhaps 40 times a winter. Storage at the ridge would be 50 MW for four hours at 85 per cent round trip. That moves surplus in time rather than in space. The second circuit has been on the reinforcement list for two years without a date, and commissioning runs about three years from a start. Curtailment payments recur every night like this one. Running above a thermal rating anneals the conductor, and the annealing accumulates.

ASK 12W

Decide what to do with more wind than the corridor can carry

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

power has to have a path to where it is used

VERDICT 79W

Energy that cannot reach a customer is not energy, and a corridor at its thermal limit is a hard wall rather than a preference. So some of that output has to stop, and the question is only how the system pays for it and what it buys instead. Reinforcing the corridor fixes it permanently and takes years. Storage on the ridge moves the surplus in time rather than in space, which is exactly the right shape for this problem.

TAKEAWAY 12W

Curtailment is what a network limit looks like from the generator's end.

13.4 Fifty hertz, through the evening rise

SYSTEM OPERATIONS · A PERSON · HOLD

SCENE 29W

The evening rise is starting and the frequency is drifting under it. Governors are answering, and the only thing left on the desk is the regulating set's raise-lower control.

GUIDE 62W

Hold the frequency inside the band on the wall display. The band narrows as the evening goes on, because the tolerance a system can live with at seven o'clock is not the one it can live with at peak. The raise-lower control is what you have, and every load that comes on is a mismatch that keeps pushing until generation answers it.

ASK 10W

Hold the system at fifty hertz through the evening rise.

BEHIND THE BUTTON 154W

Why frequency drifts at all. Frequency is the running balance between what is being generated and what is being drawn. A load that comes on is not a step in the frequency — it is a step in the *rate*, and the machines slow for as long as the mismatch lasts.

What the governors have already done. Droop has answered part of each step automatically and left an offset behind: that is what droop is for, and it is why the frequency settles low rather than returning. Bringing it back to fifty is the operator's job, and it is what this control is.

Why the band narrows. Early evening the system has inertia and reserve to spare. At peak, with less spinning and more load, the same excursion is closer to the settings that shed feeders — so the same skill has to be exercised more tightly on exactly the hour it is hardest.

TAKES AS READ

frequency falls when demand exceeds generation

VERDICT 164W

Every event here is a step in the rate rather than a step in the reading. A block of load coming on does not drop the frequency by twenty millihertz and stop; it makes the system lose speed and it goes on losing while the mismatch stands. That is why the raise-lower control has to be moved to a new position and left there rather than nudged and released, and it is the whole difference between operating a system and chasing a needle. Droop has already answered part of each step and left an offset — a system that settles at 49.94 has done exactly what it was designed to do — so returning to fifty is a separate action taken by somebody. And the band narrows for a reason that is not arbitrary: at peak there is less inertia spinning and less reserve behind it, so the same excursion sits closer to the frequency at which feeders start coming off on their own.

TAKEAWAY 13W

A mismatch you do not answer is not a mismatch that goes away.

13.5 The power that never arrives

LOAD & FORECASTING · A ROOM · CALLBACK · BALLPARK

SCENE 32W

Sten Lindgren, who forecasts demand, wants the loss figure before the morning call. The argument about moving generation is really an argument about waste. The yard radio is still live beside them.

GUIDE 61W

Five numbers, and two of them are the voltage and the power factor. Neither appears in a resistive loss. Ask of each number whether the heat depends on it. Heat comes from current and resistance, and it is paid once per phase. Note the square while you are choosing. Ten per cent more current is about twenty-one per cent more heat.

ASK 10W

Estimate the power lost as heat in the loaded corridor.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current through a resistance dissipates power as heat · charge, current, voltage and resistance — taken as read

VERDICT 81W

Resistive loss in one phase is the RMS current squared times that phase resistance. A balanced three-phase corridor has three such conductors, so the total is three times $I^2 R$. The square is the key. 10% more current means about 21% more heat. High transmission voltage helps because the same real power can be moved with less current. Lower current then cuts resistive loss sharply, which is why a modest overload can consume thermal margin faster than its percentage suggests.

TAKEAWAY 18W

Loss rises with the square of current, which is why a small overload costs more than it looks.

END OF THE DAY 18W

Dispatch orders plant by cost, and the value of certainty is what stops that being the whole story.

The last morning it can be changed

PLAN CARD 127W

Monday, and the corridor decision comes back for the last time. The outage window that would let a second circuit be strung this year closes at close of business, and if it closes the next 1 is 18 months out. June Farrow has the corrected thermal record, which says the corridor has spent more hours over limit than anybody thought. Amira Haddad has a storage quote that expires with the window. Dolores Reyes has a board that will not approve both. Today you work out what the corrected record does to the corridor's remaining life, decide what the money buys, and say what the decision commits the system to for the next 18 months. Last week this was a preference. This morning it is the last morning.

BRIEFING 16W

The reinforcement decision from last week has a date on it, and the date is today.

WHAT YOU DECIDE 11W

Commit the corridor to a course, knowing the window closes today.

14.1 Hours over limit, added up

METERING & STANDARDS · A ROOM · CHOICE

SCENE 30W

Farrow's corrected record has the corridor above continuous rating for about 90 hours over 3 years. She wants to know whether that is a maintenance note or a design problem.

GUIDE 66W

Four options, and they differ in whether the damage is reversible. Ask of each what happens when the conductor cools. Above about ninety-five degrees the aluminium's grain structure relaxes, and the strength it loses stays lost. That is why the hours are added up over three years rather than judged one event at a time. Nothing fails today, which is what makes this easy to file.

ASK 12W

What do 90 accumulated hours above continuous rating mean for the conductor?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

time spent above a thermal limit accumulates rather than resetting

VERDICT 80W

Annealing is not a state the metal comes back from. Above about 95 degrees, aluminium's grain structure begins to relax, and the tensile strength it loses stays lost when it cools. The effect accumulates in hours-above-temperature, which is why the number is added up over three years rather than judged one event at a time. The consequence is not a failure today. It is a span that fails at a wind loading the design allowed for, some years from now.

TAKEAWAY 13W

Thermal damage integrates over time, so 90 scattered hours count as 90 hours.

14.2 What the window is worth

TRANSMISSION & PROTECTION · A PERSON · SCIENCETANK

SCENE 38W

Two options, one budget, one afternoon. Novak points out that only one of them still exists tomorrow, and Haddad points out that only one of them fixes the corridor. The event board is still showing the overnight state.

GUIDE 87W

The outage window closes at close of business and the next is 18 months out. The corrected record has the corridor above continuous rating for about 90 hours over three years. The storage quote expires with the window and does not change a thermal rating. Curtailment on nights like last Friday runs about 40 nights a winter. The review Farrow would want takes three weeks. The board will not approve both the storage and the circuit, and nobody who has read the corrected record has disputed it.

ASK 7W

Commit the money before the window closes

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

an option with an expiry date is worth more than the same option without one

VERDICT 71W

This is not the same question as last week, because one option now expires. Reinforcement is the only choice that removes the constraint rather than working around it. Today is the last day it can be started this year — so its cost includes 18 months of curtailment, 90 more hours of annealing. Every evening the room has to hold a corridor it has now measured as weaker than it thought.

TAKEAWAY 14W

When one path closes today, its value includes everything the eighteen-month delay would cost.

14.3 The 18 months you just bought

SYSTEM OPERATIONS · A ROOM · PROTOCOL

SCENE 31W

Reyes wants the consequences written down before the decision is signed, so that the next shift inherits a plan rather than a mood. The room is waiting for a written instruction.

GUIDE 69W

Four things on the left, and what the decision does to each on the right. Pair them by asking what changes today and what changes only when the circuit exists. A build is a date, not relief. Between now and then the corridor is as weak as it was this morning, and now known to be weaker. One of these is closed so a shift can act at 16:40.

ASK 8W

Say what the decision commits the room to

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a decision to build something is also a decision about the years before it exists

VERDICT 70W

A build is not relief. It is a date. Between today and that date the corridor is exactly as weak as it was this morning, and now known to be weaker. The room inherits three years of damage and no new capacity. So the emergency rating is recalculated on the corrected sensor. The curtailment nights are budgeted rather than absorbed. The contingency case is run at the corridor's real strength.

TAKEAWAY 22W

Committing to a fix commits you to running the system without it until it arrives, and that has to be planned too.

14.4 Four things that have to agree before the breaker closes

GENERATION • GENERATION HALL • A ROOM • CALLBACK • SEQUENCE

SCENE 32W

Haddad has a machine at rest and a system running at nominal. She wants the sequence written down before anybody is standing at the panel. The next operating call is minutes away.

GUIDE 61W

All four of these happen at the panel, so this is not a matter of what is available. Ask of each what mismatch it removes, and what that mismatch would do if the breaker closed with it still there. A speed difference and a voltage difference cause different damage. One of these four is the consequence of the others being right.

ASK 10W

Put the steps of synchronising in the order physics requires

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a generator joined to a live system is forced to turn at the system's speed • aC waveforms: frequency, period and phase — taken as read • faraday's law and electromagnetic induction — taken as read

VERDICT 80W

Each step removes a different mismatch before the breaker closes. Match frequency first, because a speed difference forces a violent torque when the machine is tied to the grid. Match voltage next to avoid a large reactive current. Then wait for phase angle to approach zero as the two waveforms drift past each other. The breaker closes only when frequency, voltage and phase are all inside their allowed windows. Closing earlier turns a synchronization error into electrical and mechanical stress.

TAKEAWAY 11W

Speed, then voltage, then phase angle, and only then the breaker.

A reversible decision is a cheap one, and the moment it stops being reversible is the moment it needs the whole argument.

What we know, and how well

BEFORE THE DAY — TAG

Catch Whitlock before she drives off

Whitlock, the assistant operator, has the only set of relay settings that were never filed, and she is walking to her van. Once she is off site they are a week away. Catch her up before she gets there, because the whole afternoon rests on a number that currently exists in one person's notebook.

PLAN CARD 119W

Friday, and the fortnight goes into a report. Dolores Reyes will not sign a document that states everything at the same confidence, because a reader who cannot tell a measurement from an inference will act on both the same way. June Farrow has the evidence file. Ravi Lindgren wants his forecast width in as a range rather than a number. Piotr Novak wants the sensor drift written as a finding rather than an excuse. Today you sort the fortnight's claims by how strongly each is held, put numbers on the ones that deserve them, and decide which of the emergency's changes should become permanent practice. This is the part of the fortnight that will still matter in a year.

BRIEFING 18W

The event report has to state what is established, what is inferred and what is still a guess.

WHAT YOU DECIDE 15W

Say how strongly each of the fortnight's claims is held, and what outlives the emergency.

15.1 Measured, inferred, assumed

METERING & STANDARDS · A ROOM · ATTEST

SCENE 32W

June Farrow, the metering engineer, has four statements ready for the final report and only two verification slots left before sign-off. Some claims are already backed. Others are still inference or suspicion.

GUIDE 57W

Four statements are ready for the report and you have two verification slots. Open each claim and read what already backs it: some are measured, some are inferred from something measured, and some are suspicion written confidently. Hold what the backing does not support, and spend the checks where a wrong claim would do the most damage.

ASK 11W

Which claims deserve the limited verification budget before they become findings?

BEHIND THE BUTTON 149W

The three categories, and why they read alike. A measured claim points at a record. An inferred one is a chain of reasoning from a record, which is only as good as its weakest step. A suspicion has neither, and in a report all three appear as declarative sentences in the same typeface.

Why consequence rather than doubt decides the spend. The least-supported claim is not automatically the one to check. A weakly backed statement nobody will act on costs little; a well-worded inference that will drive an equipment replacement costs a great deal if it is wrong.

What holding a claim means. Not deleting it — marking it as not yet supported, so the report says what it can stand behind and what it cannot. A finding is a claim the author is willing to be wrong about in public, which is a higher bar than believing it.

TAKES AS READ

a report claim can be measured, inferred or still assumed · verification time is limited before the report closes

VERDICT 81W

The report contains claims at different evidence levels. The sensor offset is critical but already backed by two independent methods and a firmware date. Permanent conductor-strength loss is plausible metallurgy, yet no sample has tested it. The storage result is measured but not central to the cause finding. The proposed relay cause is both critical and still unproved. With only two verification slots, checking everything is impossible. Spend them on the critical claims whose present evidence cannot support a final finding.

TAKEAWAY 23W

A signed claim is still only as strong as the evidence behind it; verification should go first to critical claims that remain unbacked.

15.2 The number that has to be a range

LOAD & FORECASTING · A PERSON · CHOICE

SCENE 30W

Lindgren's winter peak goes into the report. He wants the width in the sentence, and Reyes wants to know what the width is for. The control-room phone is already ringing.

GUIDE 61W

Four reasons for a range, and they differ in what the width is for. Ask of each whether it is about blame, measurement, or planning. Reserve is not held against the expected peak. It is held against the amount by which the peak might exceed it, and that is the width. A single figure deletes the only quantity a planner needs.

ASK 16W

Why does the winter peak go in as a range rather than as a single figure?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a forecast is a distribution rather than a single value · charge, current, voltage and resistance — taken as read

VERDICT 81W

Reserve is not held against the expected peak. It is held against the amount by which the peak might exceed it. That is a property of the width and not of the centre. A report that states a single megawatt figure has deleted the only quantity a planner needs. The planner then invents one — usually badly, and usually optimistically. The width is also the honest part: it says which conditions the model was calibrated for and which it was not.

TAKEAWAY 16W

A prediction is only usable by a planner if it says how wrong it might be.

15.3 What outlives the emergency

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 34W

Four things the room started doing under pressure. Reyes asks which of them are worth their cost on an ordinary Monday, when nothing is going wrong. A printed network diagram lies on the desk.

GUIDE 55W

All four changes are defensible and all four cost something every day. Ask of each whether it would have caught this fortnight's failure, and whether the room will still be doing it in March. The sensor's offset survived months because nothing was compared against it. More frequent reports of a wrong number are not information.

ASK 10W

Which of the fortnight's changes is most worth keeping permanently?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a practice adopted under pressure has a cost that only shows up later

VERDICT 83W

One of these would have caught the whole thing. The sensor's offset survived for months because nothing was ever compared against it, and an independent check costs a survey and an afternoon. Elevated reserve is expensive every night and only useful on the rare ones. That is what a width is for deciding rather than a fortnight. More frequent loading reports would not have helped at all, because the readings were wrong rather than late. More of a wrong number is not information.

TAKEAWAY 23W

Keep the practices that would have caught the fault early, and drop the ones that only paid off because the fortnight was abnormal.

15.4 What actually limits a conductor

TRANSMISSION & PROTECTION · A ROOM · CALLBACK · PROTOCOL

SCENE 35W

Novak is asked to allow more current down a circuit for the afternoon. He wants each reason for saying no matched to the thing it is actually about. The shift log is open beside them.

GUIDE 63W

Four observations, and each is limited by a different thing. Ask of each whether it is set by current, by voltage, or by geometry. Heat comes from current and does not care about voltage. Insulation cares about voltage and not about current. A volt drop is a customer problem. One of these four is where a heat limit turns into a safety one.

ASK 7W

Match each observation to what it constrains

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a conductor warms up as current passes through it · electrical power and energy over time — taken as read

VERDICT 83W

Two different limits get confused. Heat comes from current, and it does not care what voltage the conductor is at. The insulation and the clearances care about voltage, and they do not care what current is flowing. So a circuit can be at nominal voltage and still be over its thermal limit, which is the case in front of the room. Volt drop is a third thing again, a consequence of current, and it is a customer problem rather than a safety one.

TAKEAWAY 14W

Current sets the heat, voltage sets the insulation, and the two limits are separate.

END OF THE DAY 21W

A finding is worth what its evidence is worth, and saying so is what makes a report usable a year later.

THE CLOSING CARD

The valley was back inside three hours and the report went out on the Friday, with the findings ranked by the evidence under them and the cause of the trip written down as an open question. The corridor sensor reads true now. The second circuit is strung and energising next autumn, which means eighteen more months of the corridor you have been holding all fortnight, at the strength you now know it has.

What it cost: seven hundred and fifty megawatt-hours never delivered, a fifth of an island shed to hold its frequency, and ninety hours of a conductor quietly losing strength that nobody had added up. What is unfinished: the trip has no proven cause, the store has been measured once, and the practice that would have caught all of it — check any instrument a decision rests on against something independent — is a paragraph in a report until somebody on nights makes it a habit.

Card text only — no options, boards or answers. 12,789 words of card prose across 61 stops. Generated from themes/blackout by tools/make-cardbook.mjs.